Northstar Secure Corporate Network

Network Design, Security Implementation and Validation in Cisco Packet Tracer

Technical Documentation

Portfolio Project

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1. Project Overview

1.1 Project Background

The **Northstar Secure Corporate Network** is a Cisco Packet Tracer project designed to simulate the network infrastructure of a fictional organization, **Northstar Investigations Ltd.** The organization operates a main headquarters and a remote branch office and requires secure connectivity between employees, internal services, network infrastructure, security systems, and external resources.

The project was developed as a practical networking and cybersecurity exercise, with an emphasis on building a segmented corporate environment rather than a flat network. Separate VLANs are used for Management, Investigations, IT, Servers, Branch Staff, Guest Wi-Fi, and CCTV/Security systems, with access between these networks controlled according to their operational requirements.

The topology also includes a simulated external network, allowing Internet-bound routing, Network Address Translation (NAT/PAT), firewall inspection, and unsolicited inbound traffic to be tested within Packet Tracer.

The project is implemented in two stages. The first version uses static client addressing to establish and validate the complete network architecture and security policy. The second version extends the same design with centralized DHCP and DHCP relay to demonstrate a more scalable approach to client address management.

Lab Environment Note: This project was developed as a controlled Cisco Packet Tracer simulation for learning and portfolio purposes. To simplify testing and ensure reproducibility, the password `password123` is intentionally used for administrative and selected service accounts. These credentials are not representative of production security practices.

1.2 Project Objectives

The primary objective of the project is to design, implement, secure, and validate a realistic multi-site corporate network using Cisco Packet Tracer. The network is intended to demonstrate how common enterprise networking technologies can be combined to provide connectivity while maintaining separation between systems with different security requirements.

The main technical objectives are:

- Design an IPv4 addressing scheme using **VLSM** for networks with different host requirements.
- Segment the corporate environment using dedicated **VLANs** for different users, services, and security functions.

- Implement **802.1Q trunking** and **router-on-a-stick** for inter-VLAN communication.
- Provide dynamic routing between Headquarters and the Branch using **OSPF**.
- Apply **ACLs** to enforce least-privilege communication between network segments.
- Isolate the **Guest Wi-Fi** network from internal corporate resources.
- Restrict the **CCTV/Security VLAN** while allowing authorized Management and IT access.
- Protect the network perimeter using an **IOS Zone-Based Firewall (ZBF)** and **PAT**.
- Restrict administrative access to network devices using **SSH and management ACLs**.
- Provide and test internal services including **DNS, HTTP, and FTP**.
- Validate the design through systematic connectivity, security, routing, and service testing.
- Extend the initial static-addressing implementation with **centralized DHCP and DHCP relay**.

1.3 Implementation Stages

The Northstar network is developed in two implementation stages. The first establishes and validates the complete network using **static client addressing**. Once the underlying architecture, routing, security policies, and services have been verified, the second stage introduces **centralized DHCP** while retaining the same network design. This approach allows the effect of the addressing change to be clearly demonstrated without redesigning the rest of the infrastructure.

1.3.1 Version 1 – Static Addressing

The first version uses manually configured IPv4 addresses for end-user devices. Static addressing provides a predictable environment during the initial implementation, making it easier to configure and troubleshoot VLANs, routing, ACLs, network services, PAT, and firewall policies.

Infrastructure components such as router interfaces, switch management interfaces, servers, the wireless access point, firewall interfaces, and CCTV/Security devices also use fixed addresses. This version serves as the validated baseline for the project.

1.3.2 Version 2 – Centralized DHCP

The second version extends the validated network by introducing centralized DHCP for appropriate client VLANs. The existing VLAN structure, routing, OSPF configuration, security policies, and perimeter design remain largely unchanged.

Because the DHCP server is located in the Servers VLAN, DHCP relay is configured on the relevant router interfaces to forward client DHCP requests across Layer 3 boundaries. Infrastructure devices and services that require predictable addresses continue to use static addressing.

2. Network Architecture

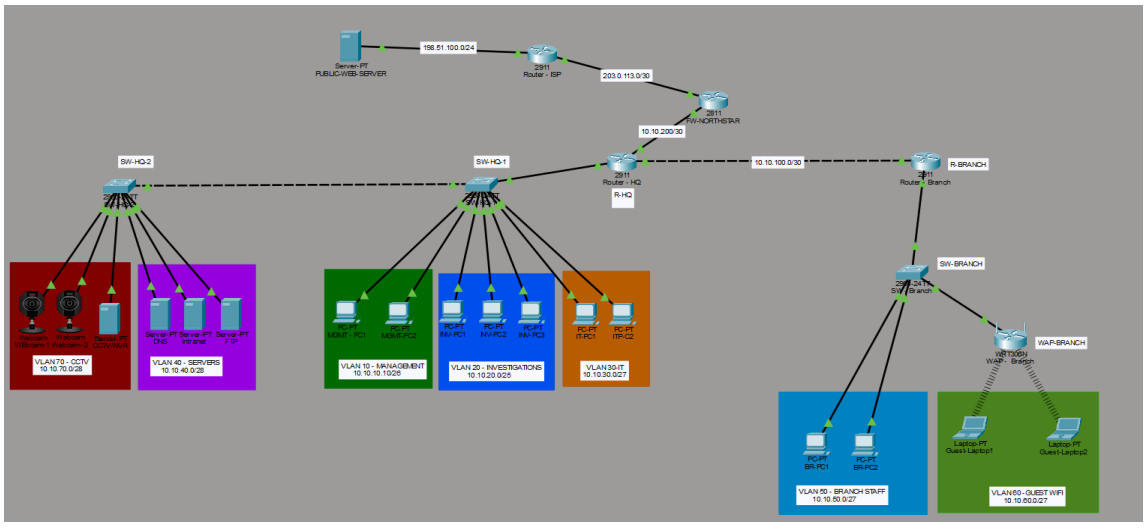
2.1 Overall Topology

The Northstar Secure Corporate Network uses a multi-site architecture consisting of a **Headquarters (HQ)** network, a **Branch** network, a perimeter security layer, and a simulated external network.

The Headquarters contains the primary corporate infrastructure and is divided into dedicated VLANs for **Management, Investigations, IT, Servers, and CCTV/Security**. Two access switches provide Layer 2 connectivity, while the HQ router performs inter-VLAN routing using an 802.1Q router-on-a-stick configuration. The Branch contains separate VLANs for **Branch Staff and Guest Wi-Fi**. A branch router provides inter-VLAN routing and connects the site to Headquarters through a dedicated point-to-point transit network. Dynamic routing between the two sites is provided by **OSPF**.

Internet-bound traffic from the corporate network is forwarded from the HQ router to **FW-NORTHSTAR**, a Cisco IOS router configured as the network perimeter firewall. The firewall provides **Zone-Based Firewall (ZBF)** inspection and **Port Address Translation (PAT)** before forwarding permitted traffic to the simulated ISP. A separate ISP router and public web server represent the external network. This provides a controlled environment in which outbound connectivity, address translation, firewall inspection, and unsolicited inbound traffic can be tested without relying on an actual Internet connection.

Figure 1. Network Topology



2.2 Device Inventory

The topology uses Cisco routers and switches together with simulated servers, end-user devices, wireless infrastructure, and security devices available in Cisco Packet Tracer. Each device has a defined role within the network architecture. Table 1 summarizes the principal infrastructure components used in the project.

Table 1. Principal Network Device Inventory

Device	Type	Location	Function
R-HQ	Cisco 2911 Router	Headquarters	Inter-VLAN routing, OSPF, and connectivity to the Branch and perimeter firewall
R-BRANCH	Cisco 2911 Router	Branch	Inter-VLAN routing for Branch networks and OSPF connectivity to Headquarters
FW-NORTHSTAR	Cisco 2811 Router	Perimeter	IOS Zone-Based Firewall, PAT, and perimeter routing
R-ISP	Cisco 2911 Router	External	Simulated Internet service provider
SW-HQ-1	Cisco 2960 Switch	Headquarters	Primary Layer 2 access and trunk connectivity
SW-HQ-2	Cisco 2960 Switch	Headquarters	Additional Layer 2 access connectivity
SW-BRANCH	Cisco 2960 Switch	Branch	Layer 2 connectivity for Branch networks
WAP-BRANCH	WRT300N	Branch	Wireless access for the Guest Wi-Fi network
DNS SERVER	Server-PT	Headquarters	Internal DNS service
HTTP SERVER	Server-PT	Headquarters	Internal corporate web service
FTP SERVER	Server-PT	Headquarters	Internal file-transfer service
CCTV-NVR	Server-PT	Headquarters	Simulated network video recorder for the CCTV/Security network
PUBLIC-WEB-SERVER	Server-PT	External	Simulated public Internet web server

End-user devices are distributed across the Management, Investigations, IT, Branch Staff, and Guest Wi-Fi networks. Two simulated IP cameras are also connected to the dedicated CCTV/Security VLAN. These endpoints are used throughout the project to verify connectivity and enforcement of the configured security policies.

2.3 VLAN and IP Addressing Plan

The Northstar network uses VLAN-based segmentation to separate users, services, and security systems according to their operational roles. Each VLAN is assigned an independent IPv4 subnet, with subnet sizes selected according to the expected number of hosts. The addressing plan uses Variable Length Subnet Masking (VLSM) to allocate address space efficiently while maintaining clear separation between network segments.

The default gateway for each VLAN is configured on the corresponding router subinterface. Headquarters VLANs are routed by R-HQ, while the Branch Staff and Guest Wi-Fi VLANs are routed by R-BRANCH.

Table 2. VLAN and IPv4 Addressing Plan

VLAN	Name / Purpose	Network	Prefix	Default Gateway	Usable Hosts
10	Management	10.10.10.0	/26	10.10.10.1	62
20	Investigations	10.10.20.0	/25	10.10.20.1	126
30	IT	10.10.30.0	/27	10.10.30.1	30
40	Servers	10.10.40.0	/28	10.10.40.1	14
50	Branch Staff	10.10.50.0	/27	10.10.50.1	30
60	Guest Wi-Fi	10.10.60.0	/27	10.10.60.1	30
70	CCTV/Security	10.10.70.0	/28	10.10.70.1	14

2.4 VLSM and Transit Networks

In addition to the user and service VLANs, dedicated transit networks are used for point-to-point communication between routing and perimeter devices. Small /30 subnets are used for these links because they provide two usable IPv4 addresses, which is sufficient for a connection between two network devices.

The use of different prefix lengths across the topology demonstrates VLSM principles: larger subnets are allocated to networks with greater host requirements, while smaller subnets are used where only a limited number of addresses are required.

Table 3. Transit and External Network Addressing

Network	Prefix	Purpose	Address Assignment
10.10.100.0	/30	HQ ↔ Branch transit	R-HQ: 10.10.100.1; R-BRANCH: 10.10.100.2

10.10.200.0	/30	HQ ↔ Perimeter firewall	R-HQ: 10.10.200.1; FW-NORTHSTAR: 10.10.200.2
203.0.113.0	/30	Firewall ↔ Simulated ISP	R-ISP: 203.0.113.1; FW-NORTHSTAR: 203.0.113.2
198.51.100.0	/24	Simulated external network	R-ISP: 198.51.100.1; PUBLIC-WEB-SERVER: 198.51.100.10

The 203.0.113.0/24 and 198.51.100.0/24 address blocks are reserved for documentation and example purposes. Their use in this project clearly distinguishes the simulated external environment from the private corporate addressing space. The public web server provides a controlled destination for testing Internet-bound routing, PAT, firewall inspection, and inbound security behavior.

Internal corporate addressing is based on the 10.10.0.0/16 private address space. The perimeter firewall therefore uses a summarized route for 10.10.0.0/16 toward R-HQ, allowing the internal networks to be reached through a single static route rather than maintaining a separate route for each VLAN.

Table 4. Configured Trunk Links

Trunk Link	Allowed VLANs	Purpose
SW-HQ-1 ↔ R-HQ	10, 20, 30, 40, 70	Carries HQ VLANs for inter-VLAN routing
SW-HQ-1 ↔ SW-HQ-2	10, 40, 70	Extends required HQ VLANs between switches
SW-BRANCH ↔ R-BRANCH	50, 60	Carries Branch Staff and Guest Wi-Fi VLANs for inter-VLAN routing

3. Switching and Routing

3.1 VLANs and Trunking

LANs are used throughout the Northstar network to create separate Layer 2 broadcast domains for users and systems with different operational and security requirements. This segmentation provides the foundation for the access-control policies implemented later in the project.

At Headquarters, VLANs 10, 20, 30, 40, and 70 represent Management, Investigations, IT, Servers, and CCTV/Security respectively. At the Branch site, VLAN 50 is assigned to Branch Staff and VLAN 60 to Guest Wi-Fi.

End-device switch ports are configured as access ports and assigned to the appropriate VLAN. Connections that must carry traffic for multiple VLANs are configured as **IEEE 802.1Q trunks**. The trunk between SW-HQ-1 and R-HQ carries the Headquarters VLANs to the router for inter-VLAN routing, while the trunk between SW-BRANCH and R-BRANCH performs the equivalent function at the Branch site.

Trunk links are also restricted to carry only the VLANs required on each connection. This reduces unnecessary VLAN propagation and provides a more controlled Layer 2 configuration.

3.2 Router-on-a-Stick

Inter-VLAN routing is implemented using a **router-on-a-stick** design at both the Headquarters and Branch sites. In this configuration, a single physical router interface carries traffic for multiple VLANs through an 802.1Q trunk, while logical subinterfaces provide a Layer 3 gateway for each VLAN.

On R-HQ, subinterfaces are configured for VLANs 10, 20, 30, 40, and 70. Each subinterface uses encapsulation dot1Q with the corresponding VLAN ID and is assigned the default gateway address for that subnet. R-BRANCH uses the same approach for VLANs 50 and 60.

For example, the Management VLAN is routed through the following R-HQ subinterface:

```
interface GigabitEthernet0/0.10

 encapsulation dot1Q 10

 ip address 10.10.10.1 255.255.255.192
```

Traffic sent by a Management host to a device in another subnet is forwarded to 10.10.10.1. The router removes the incoming Layer 2 framing, makes a Layer 3 routing decision, and forwards the packet through the appropriate interface or VLAN subinterface according to its routing table and configured security policies.

This design provides inter-VLAN connectivity without requiring a separate physical router interface for every VLAN and is well suited to the scale and educational objectives of the Packet Tracer environment.

3.3 OSPF

Open Shortest Path First (OSPF) is used as the dynamic routing protocol between the Headquarters and Branch routers. OSPF allows the routers to exchange information about their directly connected networks automatically, avoiding the need to maintain individual static routes between the two sites.

Both routers participate in **OSPF area 0** across the 10.10.100.0/30 transit network. R-HQ advertises the Headquarters VLAN networks, while R-BRANCH advertises the Branch Staff and Guest Wi-Fi networks.

The VLAN-facing interfaces are configured as **passive interfaces**. Their networks are still advertised through OSPF, but OSPF neighbor relationships are not attempted on user-facing networks. Only the transit interface between R-HQ and R-BRANCH is used to establish an OSPF adjacency.

```
router ospf 1
 router-id 1.1.1.1
 network 10.10.10.0 0.0.0.63 area 0
 network 10.10.20.0 0.0.0.127 area 0
 network 10.10.30.0 0.0.0.31 area 0
 network 10.10.40.0 0.0.0.15 area 0
 network 10.10.70.0 0.0.0.15 area 0
 network 10.10.100.0 0.0.0.3 area 0
 passive-interface default
 no passive-interface GigabitEthernet0/1
 default-information originate
```

R-HQ also originates a default route into OSPF. As a result, R-BRANCH does not require a separately configured static default route toward the Internet. Instead, it learns the default route dynamically from R-HQ and forwards external traffic across the HQ–Branch transit link.

Successful OSPF operation was confirmed by establishing a **FULL neighbor adjacency** between R-HQ and R-BRANCH and verifying that remote VLAN networks and the OSPF-learned default route appeared in the routing tables.

3.4 Default and Static Routing

Dynamic routing is used between Headquarters and the Branch, while static and default routes are used at the network perimeter. This combination keeps internal multi-site routing dynamic while providing a simple and predictable path toward the simulated Internet.

R-HQ uses a static default route pointing to the inside interface of FW-NORTHSTAR. Traffic for destinations that are not present in the internal routing table is therefore forwarded to the perimeter firewall.

```
ip route 0.0.0.0 0.0.0.0 10.10.200.2
```

FW-NORTHSTAR uses its own default route toward R-ISP:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

For return traffic toward the corporate network, the firewall uses a summarized static route covering the internal 10.10.x.x addressing space:

```
ip route 10.10.0.0 255.255.0.0 10.10.200.1
```

The 10.10.0.0/16 summary avoids the need to configure an individual static route on the firewall for every internal VLAN. Although the summary also encompasses currently unused address space, it provides a simple and scalable route toward R-HQ for the internal addressing plan.

The directly connected 10.10.200.0/30 network is not affected by the broader /16 route. Routers use **longest-prefix matching**, so the more specific directly connected /30 route is preferred whenever the destination belongs to that network.

Together, these routing mechanisms create the external traffic path **Branch/HQ → R-HQ → FW-NORTHSTAR → R-ISP → external network**. R-HQ also advertises its default route through OSPF, allowing R-BRANCH to learn the Internet path dynamically.

4. Network Security

4.1 Network Segmentation and ACLs

Network segmentation in the Northstar environment is enforced through a combination of VLAN separation and router-based **Access Control Lists (ACLs)**. VLANs create separate Layer 2 networks, while ACLs control which types of traffic are permitted to cross Layer 3 boundaries.

The access-control design follows the **principle of least privilege**. Networks are permitted to communicate only with the systems and services required for their intended function. Extended ACLs are generally applied close to the traffic source, allowing unwanted traffic to be filtered before it traverses other parts of the network.

A key example is the Investigations VLAN. Investigation workstations require access to selected internal services but do not require unrestricted access to other corporate networks. The ACL therefore permits DNS queries to the internal DNS server, HTTP/HTTPS access to the internal web server, and FTP access to the designated FTP server. Broader access to protected internal VLANs is denied, while permitted Internet-bound traffic can continue toward the perimeter.

```
ip access-list extended INVESTIGATIONS-IN

 permit udp 10.10.20.0 0.0.0.127 host 10.10.40.2 eq 53

 permit tcp 10.10.20.0 0.0.0.127 host 10.10.40.2 eq 53

 permit tcp 10.10.20.0 0.0.0.127 host 10.10.40.3 eq 80

 permit tcp 10.10.20.0 0.0.0.127 host 10.10.40.3 eq 443

 permit tcp 10.10.20.0 0.0.0.127 host 10.10.40.4 eq 21

 permit tcp 10.10.20.0 0.0.0.127 host 10.10.40.4 eq 20
```

The order of ACL entries is significant because Cisco ACLs are evaluated sequentially from top to bottom and processing stops at the first matching entry. Service-specific permit statements must therefore appear before broader deny statements that would otherwise block the same traffic.

Similar policies are used for the Branch, Guest Wi-Fi, and CCTV/Security networks. The exact restrictions vary according to the role of each network and are described in the following subsections.

4.2 Guest Wi-Fi Isolation

The Branch Guest Wi-Fi network is placed in a dedicated **VLAN 60** to separate untrusted wireless clients from corporate systems. Guest devices require external connectivity but have no operational requirement to access internal Northstar resources.

An extended ACL is therefore applied to traffic entering R-BRANCH from the Guest VLAN. The policy explicitly denies access from the 10.10.60.0/27 network to the internal Management, Investigations, IT, Servers, Branch Staff, and CCTV/Security networks. Traffic that does not match these internal destinations is permitted to continue toward the network perimeter.

This creates a simple security boundary: guest clients can reach permitted external resources through the corporate Internet connection while remaining isolated from internal systems.

```
ip access-list extended GUEST-IN
 deny ip 10.10.60.0 0.0.0.31 10.10.10.0 0.0.0.63
 deny ip 10.10.60.0 0.0.0.31 10.10.20.0 0.0.0.127
 deny ip 10.10.60.0 0.0.0.31 10.10.30.0 0.0.0.31
 deny ip 10.10.60.0 0.0.0.31 10.10.40.0 0.0.0.15
 deny ip 10.10.60.0 0.0.0.31 10.10.50.0 0.0.0.31
 deny ip 10.10.60.0 0.0.0.31 10.10.70.0 0.0.0.15
 permit ip 10.10.60.0 0.0.0.31 any
```

WAP-BRANCH operates as a wireless access point rather than as an independent routing or DHCP device. This keeps routing and security enforcement on R-BRANCH and prevents the wireless device from creating a separate, uncontrolled network boundary.

4.3 CCTV/Security VLAN

CCTV and security devices are placed in a dedicated **VLAN 70** to isolate them from general-purpose corporate endpoints. This reflects the principle that security and IoT-type devices should not automatically receive unrestricted access to user networks or external resources.

The simulated CCTV environment consists of two IP cameras and a CCTV-NVR. Management and IT users are permitted to access the NVR for administrative and monitoring purposes, while devices within the CCTV/Security VLAN are prevented from initiating arbitrary routed connections to other networks.

Because ACLs are stateless, specific return-traffic rules are included to allow the NVR to respond to authorized connections initiated from the Management and IT networks. Other traffic originating from VLAN 70 is denied.

```
ip access-list extended CCTV-IN
 permit tcp host 10.10.70.4 10.10.10.0 0.0.0.63 established
 permit tcp host 10.10.70.4 10.10.30.0 0.0.0.31 established
 permit icmp host 10.10.70.4 10.10.10.0 0.0.0.63 echo-reply
 permit icmp host 10.10.70.4 10.10.30.0 0.0.0.31 echo-reply
 deny ip 10.10.70.0 0.0.0.15 any
```

VLAN 70 is also excluded from the PAT configuration used for general Internet access. The project does not expose the NVR or cameras directly to the simulated Internet through port forwarding.

In a production environment, remote CCTV access should instead be provided through an authenticated **VPN** or an appropriately secured outbound cloud-mediated solution. Direct public exposure of CCTV management interfaces would unnecessarily increase the attack surface.

4.4 Zone-Based Firewall

Perimeter security is provided by **FW-NORTHSTAR**, a Cisco 2811 router configured with the Cisco IOS **Zone-Based Firewall (ZBF)** feature. The firewall separates the trusted corporate network from the simulated external network and provides stateful traffic inspection between the two security zones.

Two security zones are defined: **INSIDE** for the interface facing R-HQ and **OUTSIDE** for the interface facing R-ISP. A zone pair controls traffic initiated from INSIDE toward OUTSIDE. The configured policy permits and inspects HTTP and HTTPS traffic.

Stateful inspection allows return traffic associated with an authorized outbound session to pass back through the firewall without requiring a separate inbound permit rule for each internal host. Unsolicited traffic initiated from the OUTSIDE zone toward the corporate network is not permitted by the configured policy.

```
zone security INSIDE

zone security OUTSIDE

class-map type inspect match-any ALLOWED_TRAFFIC

 match protocol http

 match protocol https

policy-map type inspect WEB_POLICY

 class type inspect ALLOWED_TRAFFIC

  inspect

zone-pair security INSIDE_TO_OUTSIDE source INSIDE destination OUTSIDE

 service-policy type inspect WEB_POLICY
```

The firewall interfaces are assigned to their corresponding zones, with the internal interface connected to the 10.10.200.0/30 transit network and the external interface connected to the 203.0.113.0/30 simulated ISP network.

```
interface FastEthernet0/0

 zone-member security INSIDE

interface FastEthernet0/1

 zone-member security OUTSIDE
```

The ZBF configuration complements, rather than replaces, the internal ACL-based segmentation. Internal router ACLs control communication between corporate networks, while the perimeter firewall controls stateful communication between the corporate environment and the simulated external network. This provides multiple layers of traffic control within the overall design.

4.5 Port Address Translation (PAT)

Port Address Translation (PAT) is configured on FW-NORTHSTAR to allow multiple internal hosts to access the simulated external network while sharing the firewall's external IPv4 address. Internal private addresses are translated to the address assigned to the firewall's outside interface, with individual sessions distinguished by transport-layer port numbers.

The firewall defines its corporate-facing interface as the NAT inside interface and its ISP-facing interface as the NAT outside interface.

```
interface FastEthernet0/0

 ip nat inside

interface FastEthernet0/1

 ip nat outside
```

An access list identifies the internal networks that are eligible for translation. PAT is enabled using the overload keyword.

```
ip access-list standard NAT_ELIGIBLE

 permit 10.10.10.0 0.0.0.63

 permit 10.10.20.0 0.0.0.127

 permit 10.10.30.0 0.0.0.31

 permit 10.10.50.0 0.0.0.31

 permit 10.10.60.0 0.0.0.31

ip nat inside source list NAT_ELIGIBLE interface FastEthernet0/1 overload
```

Only networks requiring external access are included in the PAT eligibility list. The Servers VLAN and CCTV/Security VLAN are intentionally excluded, preventing them from receiving general Internet access through this translation policy.

PAT and the Zone-Based Firewall perform different functions. PAT translates internal addresses for external communication, while ZBF determines whether traffic is permitted and provides stateful inspection. Using both mechanisms provides address translation together with perimeter traffic control.

4.6 Secure Device Management

Administrative access to network infrastructure is restricted and protected using **SSH** rather than insecure plaintext remote-management protocols such as Telnet. Local administrator accounts, encrypted credentials, login banners, and inactivity timeouts are configured on managed network devices.

Access to the VTY lines is further restricted using a standard management ACL. Only hosts originating from the dedicated **Management VLAN (10.10.10.0/26)** are permitted to establish remote SSH management sessions.

```
username admin privilege 15 secret password123

ip domain-name northstar.local

ip ssh version 2
```

```
ip access-list standard MGMT-SSH
 permit 10.10.10.0 0.0.0.63
 deny any
line vty 0 4
 access-class MGMT-SSH in
 login local
 transport input ssh
 exec-timeout 5 0
```

Similar management restrictions are applied to the switching infrastructure. This separates the **management plane** from normal user traffic: a device may be reachable for ordinary network communication without automatically being permitted to administer routers, switches, or the perimeter firewall.

Unauthorized SSH attempts from non-management networks were tested and rejected, while authorized Management VLAN hosts were able to establish SSH sessions successfully.

The credentials used in this Packet Tracer environment are intentionally simplified for demonstration purposes, as described in Section 1.1. A production deployment would require unique strong credentials and appropriate enterprise authentication and credential-management mechanisms.

5. Network Services

5.1 DNS and Web Services

The Servers VLAN (10.10.40.0/28) hosts the primary internal services used by the Northstar network. These services are assigned static IPv4 addresses so that they remain consistently reachable by clients and can be referenced reliably in ACLs and DNS records.

The internal DNS server is configured at 10.10.40.2 and provides name resolution for selected Northstar services. The following DNS records are configured:

Table 5. Internal DNS Records.

DNS Name	IPv4 Address	Purpose
intranet.northstar.local	10.10.40.3	Internal web server
files.northstar.local	10.10.40.4	Internal FTP server
nvr.northstar.local	10.10.70.4	CCTV NVR

5.2 FTP Service

A dedicated FTP server is configured in the Servers VLAN at 10.10.40.4. The service provides a simple internal file-transfer platform and is used to demonstrate service-specific access control between network segments.

An FTP user account is configured for testing purposes, and successful file transfer was verified from an authorized client. The Investigations VLAN ACL specifically permits FTP control and data traffic to this server while restricting broader access to the Servers VLAN.

The FTP implementation is intended for laboratory demonstration only. Traditional FTP transmits credentials and data without encryption and would not be appropriate for sensitive information in a production environment. A real deployment would use a secure alternative such as **SFTP or another encrypted file-transfer mechanism**.

5.3 CCTV NVR Simulation

The CCTV/Security environment includes two simulated IP cameras and a dedicated **CCTV-NVR** located in VLAN 70. The NVR uses the static address 10.10.70.4 and represents the centralized monitoring system for the security network.

Because Cisco Packet Tracer does not provide the functionality of a complete production Network Video Recorder, a Server-PT device is used to simulate the NVR. Its web service provides a simple management interface that can be reached by authorized Management and IT hosts.

The NVR is also registered in the internal DNS service as nvr.northstar.local, allowing authorized users to access it by hostname rather than only by IPv4 address.

The simulation is primarily used to demonstrate **network segmentation and controlled administrative access**. The project does not attempt to emulate video recording, storage, camera authentication, or other functions of a real CCTV platform.

5.4 Simulated Internet

The external portion of the topology simulates Internet connectivity without requiring access to a real external network. **R-ISP** represents the Internet service provider, while **PUBLIC-WEB-SERVER** represents a publicly accessible Internet service.

FW-NORTHSTAR connects to R-ISP through the 203.0.113.0/30 network. The public server is located on the separate 198.51.100.0/24 external network and uses 198.51.100.10 as its IPv4 address.

This environment provides a controlled destination for testing the complete outbound traffic path, including internal routing, PAT, Zone-Based Firewall inspection, ISP routing, and return traffic. Authorized internal clients were able to access the public HTTP service through this path.

The external network is also used for security testing. Connections initiated from the external side toward internal corporate hosts are not permitted by the configured firewall policy, allowing the behavior of unsolicited inbound traffic to be verified.

The public-facing address ranges used in the topology are documentation networks and do not represent real Internet services.

6. Addressing Implementations

6.1 Version 1 – Static Addressing

The first implementation of the Northstar network uses **static IPv4 addressing** for all infrastructure devices, servers, security devices, and end-user hosts. This provides a predictable environment for the initial deployment and simplifies troubleshooting while VLANs, routing, ACLs, firewall policies, NAT/PAT, and network services are being configured and validated.

Table 6 summarizes the principal static endpoint addresses used in the validated baseline network.

Table 6. Static Endpoint Address Assignments

Device	VLAN	IPv4 Address	Prefix	Default Gateway
MGMT-PC1	10	10.10.10.2	/26	10.10.10.1
MGMT-PC2	10	10.10.10.3	/26	10.10.10.1
SW-HQ-1	10	10.10.10.10	/26	10.10.10.1
SW-HQ-2	10	10.10.10.11	/26	10.10.10.1
INV-PC1	20	10.10.20.2	/25	10.10.20.1
INV-PC2	20	10.10.20.3	/25	10.10.20.1
INV-PC3	20	10.10.20.4	/25	10.10.20.1
IT-PC1	30	10.10.30.2	/27	10.10.30.1
IT-PC2	30	10.10.30.3	/27	10.10.30.1
DNS SERVER	40	10.10.40.2	/28	10.10.40.1
HTTP SERVER	40	10.10.40.3	/28	10.10.40.1
FTP SERVER	40	10.10.40.4	/28	10.10.40.1
BR-PC1	50	10.10.50.2	/27	10.10.50.1
BR-PC2	50	10.10.50.3	/27	10.10.50.1
SW-BRANCH	50	10.10.50.10	/27	10.10.50.1
WAP-BRANCH	60	10.10.60.2	/27	10.10.60.1
Guest-Laptop1	60	10.10.60.3	/27	10.10.60.1
Guest-Laptop2	60	10.10.60.4	/27	10.10.60.1
Webcam1	70	10.10.70.2	/28	10.10.70.1
Webcam2	70	10.10.70.3	/28	10.10.70.1

Device	VLAN	IPv4 Address	Prefix	Default Gateway
CCTV-NVR	70	10.10.70.4	/28	10.10.70.1

Static addressing was retained for infrastructure components and services that require predictable addresses, including router interfaces, switch management interfaces, servers, the wireless access point, firewall interfaces, and CCTV/Security devices.

After configuration, the static-addressing version was tested for routing, internal service access, network segmentation, Internet connectivity, PAT, firewall behavior, and secure management access. This validated version served as the baseline for the centralized DHCP implementation.

6.2 Version 2 – Centralized DHCP

The second implementation extends the validated static-addressing network by introducing **centralized DHCP** for end-user devices. The existing VLAN structure, OSPF routing, ACL policies, firewall configuration, PAT, and network services remain unchanged.

The existing DNS server at 10.10.40.2 also provides the centralized DHCP service. Separate DHCP pools are configured for each client VLAN, allowing hosts to automatically obtain an IPv4 address, subnet mask, default gateway, and DNS server.

Infrastructure and service devices continue to use static addressing because they require predictable addresses. These include router and firewall interfaces, switch management interfaces, servers, the wireless access point, and CCTV/Security devices.

Table 7. Centralized DHCP Pools

VLAN	Pool Name	DHCP Range	Subnet Mask	Gateway	DNS
10	VLAN10-MGMT	10.10.10.20–59	255.255.255.192	10.10.10.1	10.10.40.2
20	VLAN20-INV	10.10.20.20–119	255.255.255.128	10.10.20.1	10.10.40.2
30	VLAN30-IT	10.10.30.10–29	255.255.255.224	10.10.30.1	10.10.40.2
50	VLAN50-BRANCH	10.10.50.20–29	255.255.255.224	10.10.50.1	10.10.40.2
60	VLAN60-GUEST	10.10.60.10–29	255.255.255.224	10.10.60.1	10.10.40.2

Dynamic address ranges are separated from addresses reserved for infrastructure and other statically configured devices. This prevents the DHCP service from assigning addresses already used by gateways, switches, servers, or other fixed network components.

After the DHCP pools were configured and validated, the Management, Investigations, IT, Branch Staff, and Guest Wi-Fi client devices were migrated from static addressing to DHCP. The Servers and CCTV/Security VLANs remain statically addressed.

6.3 DHCP Relay

Because the centralized DHCP server is located in VLAN 40, DHCP broadcasts generated by clients in other VLANs cannot reach the server directly. Routers normally do not forward broadcast traffic between different IP networks.

To solve this, DHCP relay is configured on the Layer 3 gateway interface of each DHCP-enabled client VLAN. The Cisco IOS ip helper-address command receives the client DHCP broadcast and forwards the request toward the centralized DHCP server at 10.10.40.2.

For example, DHCP relay for the Management VLAN is configured on R-HQ as follows:

```
interface GigabitEthernet0/0.10
 ip helper-address 10.10.40.2
```

The same principle is applied to VLANs 20 and 30 on R-HQ and VLANs 50 and 60 on R-BRANCH. This allows the Branch networks to use the same centralized DHCP server even though the server is located at Headquarters and is reached through the OSPF-routed connection.

Table 8. DHCP Relay Configuration

Router	Interface	Client VLAN	Helper Address
R-HQ	Gi0/0.10	10 – Management	10.10.40.2
R-HQ	Gi0/0.20	20 – Investigations	10.10.40.2
R-HQ	Gi0/0.30	30 – IT	10.10.40.2
R-BRANCH	Gi0/0.50	50 – Branch Staff	10.10.40.2
R-BRANCH	Gi0/0.60	60 – Guest Wi-Fi	10.10.40.2

During implementation, the inbound ACLs applied to the Investigations, Branch Staff, and Guest Wi-Fi VLANs also required explicit permission for the initial DHCP client traffic. A client requesting its first lease does not yet possess an address from its VLAN subnet, so the ACLs were updated to permit DHCP client-to-server traffic using UDP ports 68 and 67 before the remaining security rules were evaluated.

```
permit udp any eq bootpc any eq bootps
```

After these changes, clients at both Headquarters and the Branch successfully obtained their addressing information from the centralized server while the existing VLAN access restrictions remained enforced.

7. Verification and Testing

After implementation, functional and security testing was performed to verify routing, network services, segmentation, firewall policies, NAT/PAT, DHCP operation, and management access. Testing included both successful connections and intentionally blocked traffic to confirm that the configured security controls operated as designed.

7.1 Routing and Connectivity

Routing was verified between the Headquarters and Branch networks after OSPF convergence. R-HQ and R-BRANCH formed an OSPF adjacency across the 10.10.100.0/30 transit network, allowing the Branch router to dynamically learn the Headquarters networks.

R-HQ also advertises the default route into OSPF, providing the Branch with a path toward the perimeter firewall and simulated Internet. Connectivity tests confirmed successful communication between networks where permitted by the security policy.

```
show ip ospf neighbor

show ip route
```

In the centralized DHCP implementation, successful lease allocation to Branch clients also demonstrated end-to-end connectivity between the Branch VLANs and the DHCP server located in VLAN 40 at Headquarters.

7.2 ACL and Segmentation Testing

Access-control testing was performed from multiple VLANs to verify that network segmentation remained effective while authorized services were still reachable. Both permitted and denied connections were tested. The results confirmed that the ACLs enforce the intended least-privilege communication policy between the different security zones.

Table 9. ACL and Network Segmentation Test Results

Source	Destination	Test	Expected	Result
Investigations	Internal Web Server	HTTP	Allowed	**PASS**
Investigations	Management VLAN	ICMP	Blocked	**PASS**

Source	Destination	Test	Expected	Result
Branch Staff	Server VLAN	Permitted services	Allowed	**PASS**
Guest Wi-Fi	Server VLAN	ICMP	Blocked	**PASS**
Guest Wi-Fi	Public Web Server	HTTP	Allowed	**PASS**
IT	CCTV-NVR	HTTP	Allowed	**PASS**
CCTV-NVR	IT workstation	ICMP	Blocked	**PASS**
Internal Server	Public Web Server	HTTP	Blocked	**PASS**

The tests demonstrate that segmentation is enforced according to device and user roles. Investigations users can access required internal services without receiving unrestricted access to other corporate VLANs. Guest devices are isolated from internal resources while retaining permitted Internet access. The CCTV/Security network prevents devices from initiating arbitrary routed connections, while authorized Management and IT systems can access the simulated NVR.

The successful tests also confirm that introducing centralized DHCP did not bypass or weaken the existing segmentation policies.

7.3 Firewall and PAT Testing

Perimeter security testing was performed to verify both outbound Internet access and protection against unsolicited inbound traffic. FW-NORTHSTAR uses IOS Zone-Based Firewall (ZBF) inspection together with Port Address Translation (PAT).

Authorized client VLANs successfully accessed the simulated public web server at 198.51.100.10 using HTTP. The firewall dynamically inspected the permitted sessions, while PAT translated multiple private client addresses to the firewall outside address 203.0.113.2.

Figure 2. PAT translations for clients from multiple VLANs

```
FW-NORTHSTAR#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 203.0.113.2:1024   10.10.20.22:1025   198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:1025   10.10.10.20:1025   198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:1026   10.10.20.22:1026   198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:1027   10.10.60.10:1027   198.51.100.10:80   198.51.100.10:80
tcp 203.0.113.2:1028   10.10.60.10:1028   198.51.100.10:80   198.51.100.10:80

FW-NORTHSTAR#
```

The NAT translation table demonstrates PAT operation for clients in the Management (10.10.10.20), Investigations (10.10.20.22), and Guest Wi-Fi (10.10.60.10) networks. All three private

addresses are translated to the firewall's outside address (203.0.113.2), while unique TCP port numbers distinguish the individual sessions.

Additional negative testing confirmed that an external host could not initiate a connection toward an internal Management workstation. Internal servers excluded from the NAT policy were also unable to initiate general Internet access, as intended.

Table 10. Firewall and PAT Test Results

Source	Destination	Test	Expected	Result
Management client	Public Web Server	HTTP	Allowed + PAT	**PASS**
Investigations client	Public Web Server	HTTP	Allowed + PAT	**PASS**
Guest client	Public Web Server	HTTP	Allowed + PAT	**PASS**
Internal Server	Public Web Server	HTTP	Blocked	**PASS**
Public Web Server	Management client	ICMP	Blocked	**PASS**

These tests demonstrate that the perimeter configuration provides controlled outbound connectivity without exposing internal hosts to unsolicited traffic from the simulated external network.

7.4 Management Access Testing

Secure management access was tested to verify that administrative interfaces could only be reached from authorized network segments. SSH was used instead of Telnet to provide encrypted remote management.

From the Management VLAN, MGMT-PC1 successfully established an SSH session with FW-NORTHSTAR at `10.10.200.2` using the configured local administrator account. This confirmed that legitimate administrative access from the designated management network was permitted.

A second SSH connection was attempted from an Investigations workstation. The connection was refused, confirming that the management-plane ACL prevented unauthorized VLANs from remotely accessing the firewall.

Table 11. Management Access Test Results

Source	Target	Protocol	Expected	Result
Management client	FW-NORTHSTAR	SSH	Allowed	**PASS**
Investigations client	FW-NORTHSTAR	SSH	Blocked	**PASS**

These results demonstrate that user data-plane connectivity and administrative management access are controlled separately. Restricting SSH access to the dedicated Management VLAN reduces the exposure of network infrastructure to unauthorized administrative access.

8. Troubleshooting and Lessons Learned

Several issues encountered during implementation required systematic troubleshooting and configuration changes. These problems provided practical experience in packet-flow analysis, ACL processing, DHCP relay, firewall behavior, and the limitations of simulated networking environments.

8.1 ASA to IOS ZBF Migration

The original perimeter design used a Cisco ASA 5506-X firewall. During testing in Packet Tracer, inside-to-outside HTTP traffic reached the simulated Internet server and PAT translation occurred correctly. However, return traffic was not consistently permitted through the simulated ASA when using the intended subnet-based policy.

Packet-level testing showed that host-specific rules could allow the traffic, but implementing individual rules for every internal client would not represent a scalable enterprise design. Rather than weakening the security policy to accommodate the simulation behavior, the perimeter device was replaced with a Cisco IOS router configured with **Zone-Based Firewall (ZBF)** and PAT.

The replacement preserved the original security objectives: controlled outbound access, stateful inspection of permitted web traffic, address translation, separation of inside and outside security zones, and protection against unsolicited inbound traffic.

The original ASA implementation was retained separately as a troubleshooting artifact, while the IOS ZBF implementation became the final validated design.

8.2 ACL Ordering Issue

During Branch connectivity testing, Branch Staff clients were initially unable to reach services in the Server VLAN even though the required traffic was intended to be permitted.

Examination of the BRANCH-IN access control list showed that a broader deny rule was evaluated before the required permit rule. Because Cisco ACLs are processed sequentially from top to bottom and stop at the first matching entry, the traffic was denied before reaching the intended permit statement.

The ACL was corrected by placing the more specific permitted traffic before the broader deny statements. Subsequent testing confirmed that Branch clients could reach the authorized server resources while access to restricted internal VLANs remained blocked.

This demonstrated the importance of **ACL rule ordering and specificity**. A logically correct set of rules can still produce incorrect behavior when entries are placed in the wrong sequence.

8.3 DHCP Troubleshooting and Packet Tracer Limitations

The introduction of centralized DHCP created an additional troubleshooting scenario because DHCP clients initially transmit from 0.0.0.0 using UDP ports 68 and 67. Existing inbound ACLs were designed around the assigned VLAN address ranges and therefore prevented these initial DHCP requests from being processed correctly.

The affected ACLs were updated with a specific DHCP bootstrap rule before the remaining security entries:

```
permit udp any eq bootpc any eq bootps
```

A separate issue occurred on the Branch Staff VLAN. DHCP still failed after the ACL modification, although R-BRANCH could successfully reach the DHCP server at 10.10.40.2. ACL hit counters confirmed that DHCP Discover traffic was reaching the router, and Packet Tracer Simulation Mode showed the packet stopping at R-BRANCH.

Inspection of the running configuration revealed that the expected ip helper-address 10.10.40.2 command was missing from the VLAN 50 subinterface. After restoring the DHCP relay configuration, the Branch client successfully obtained a lease from the centralized DHCP server.

These troubleshooting steps demonstrated the value of testing the network layer by layer rather than immediately modifying multiple configurations. Connectivity tests, ACL counters, Simulation Mode, and inspection of the active running configuration were used to progressively isolate the problem.

Packet Tracer is highly useful for demonstrating networking concepts, but its device models do not reproduce every behavior and feature of production hardware. The ASA behavior encountered earlier in the project was therefore treated as an observed simulation limitation rather than as representative of all real Cisco ASA deployments. The final design was validated using the IOS Zone-Based Firewall implementation supported reliably by the simulation environment.

9. Conclusion

9.1 Project Outcomes

The Northstar Secure Corporate Network project demonstrates the design, implementation, security, and validation of a multi-site corporate network in Cisco Packet Tracer. The completed environment integrates VLAN-based segmentation, VLSM addressing, router-on-a-stick, dynamic

routing with OSPF, centralized network services, ACL-based access control, secure device management, wireless guest access, IOS Zone-Based Firewall protection, and PAT.

Two functional versions of the network were produced. The first uses static endpoint addressing and provides a stable baseline implementation. The second introduces centralized DHCP and DHCP relay while preserving the same routing and security architecture.

Functional and negative testing confirmed that authorized communication is permitted while restricted traffic is blocked according to the intended security policy. The project also provided practical troubleshooting experience involving ACL processing, DHCP relay, routing, NAT/PAT, and firewall behavior.

9.2 Future Improvements

The network could be extended with additional enterprise features in future versions. Potential improvements include redundant DHCP and DNS services, Layer 3 switching, redundant network paths, VPN-based remote access, centralized logging and monitoring, AAA services such as RADIUS or TACACS+, and more advanced firewall policies.

A larger implementation could also introduce high availability and additional security monitoring while preserving the segmented architecture developed in this project.

Overall, the project provides a practical foundation for understanding how routing, switching, network services, and security controls interact within a structured corporate network.